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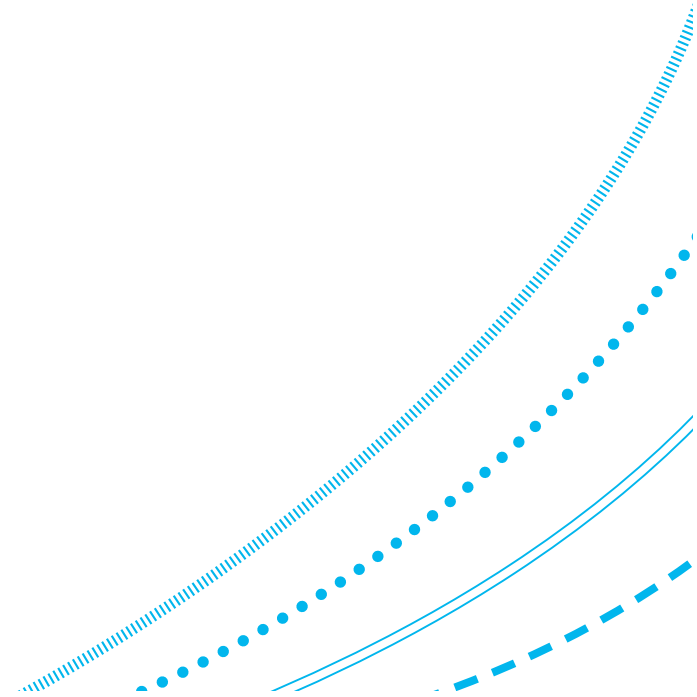
Regression Analysis

Data Mining & Visualisation
Lecture 9



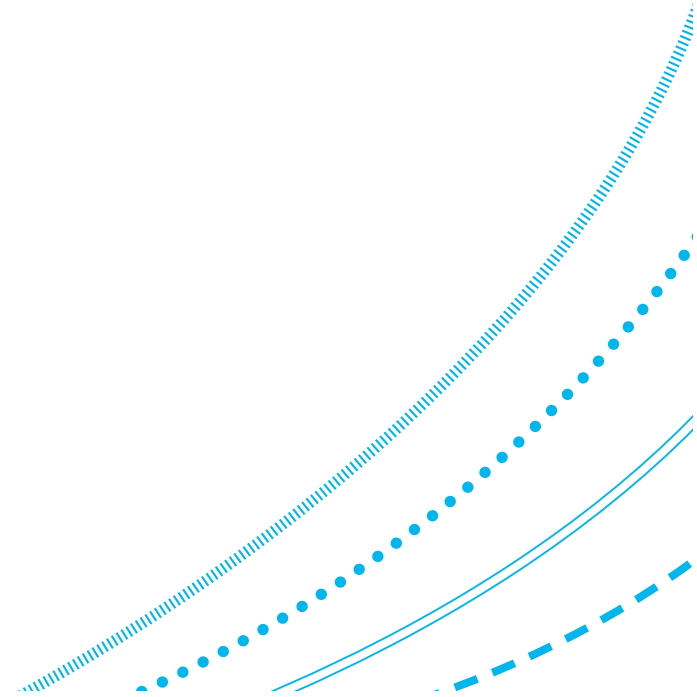
Recap

- Types of Data
 - Quantitative vs Categorical Variables
- Relationships in data
 - Correlation



Today

- Supervised Learning
- Regression Analysis
- Evaluating a Regression Model



Recap: Supervised Learning

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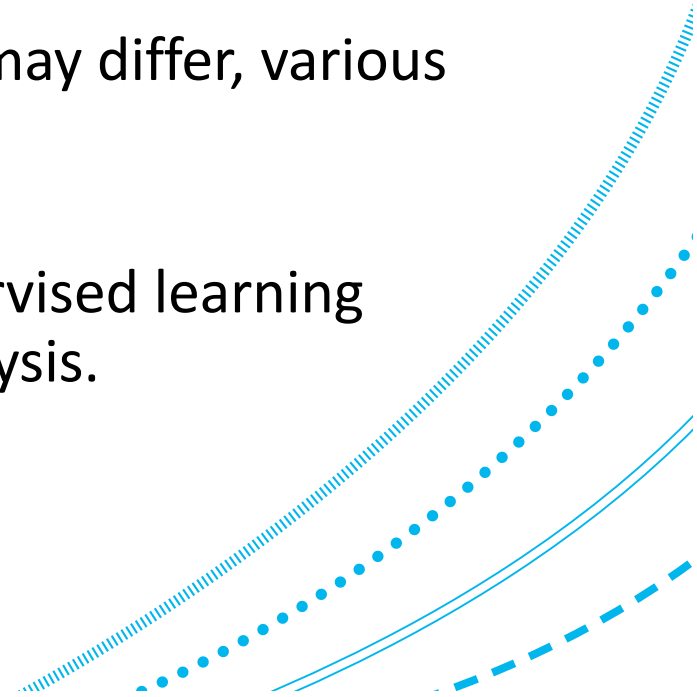


Recap: Supervised Learning

As we discussed during Lecture 2, there are a lot of overlaps between data mining and machine learning.

While the exact steps, terminology, and end-goals may differ, various aspects of today's lecture should be familiar to you.

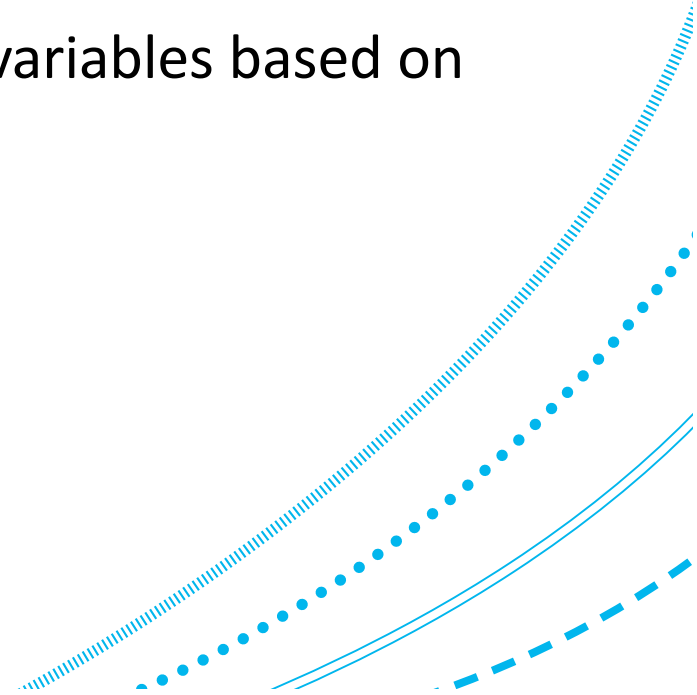
In the next few lectures, we will discuss a few supervised learning methods that can be used for data mining and analysis.



Recap: Supervised Learning

Recall that supervised learning is a type of machine learning where labelled training data is used to train models.

Our models learn the relationships between these variables based on the labelled data that we provide.



Recap: Supervised Learning

Example: Based on a housing dataset, we could train a model to map a house's size (in sq. ft.) to the value of that house.

Price (\$)	Size (Sq. ft.)
100,000	1,000
250,000	2,500
400,000	4,000
...	...

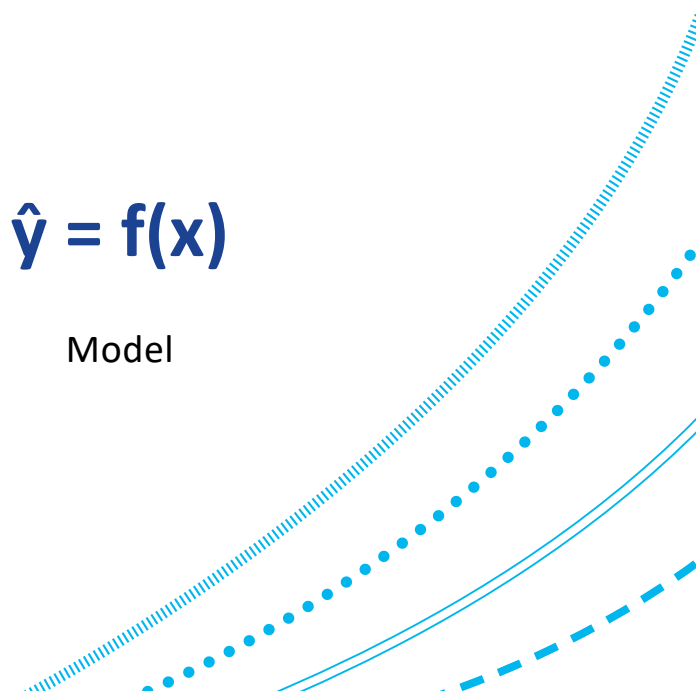
Labelled Data



Model training

$$\hat{y} = f(x)$$

Model

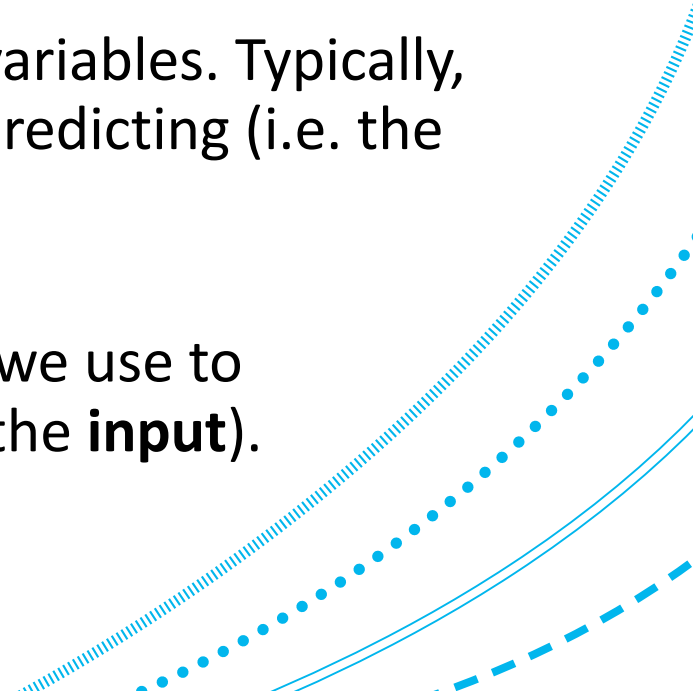


Terminology

In statistical learning, we often refer to variables as either *dependent* or *independent*.

A **dependent variable** (DV) is dependent on other variables. Typically, this is what we are interested in understanding or predicting (i.e. the **output** of our model).

An **independent variable** (IV), or predictor, is what we use to understand or predict our dependent variable (i.e. the **input**).

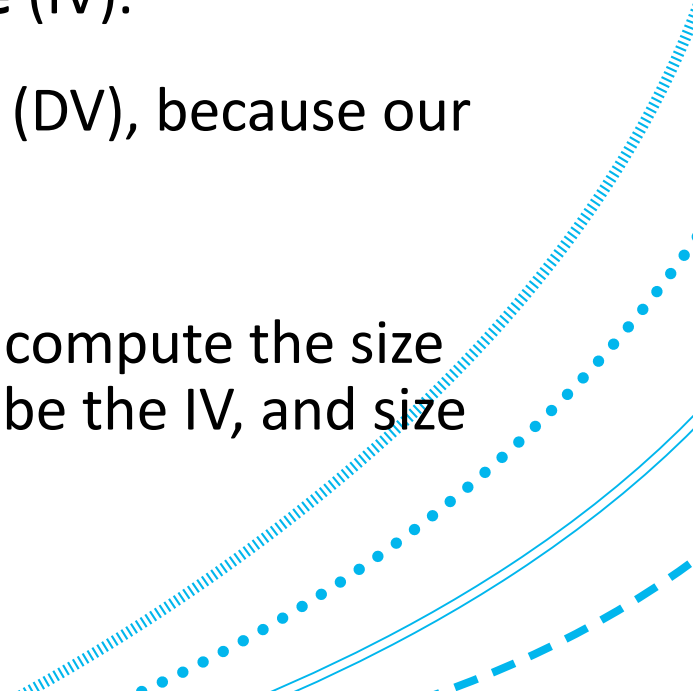


Terminology

In the example where we compute the price from the size:

- The size of the house is our **independent variable** (IV).
- The price of the house is our **dependent variable** (DV), because our model's output (price) is dependent on the size.

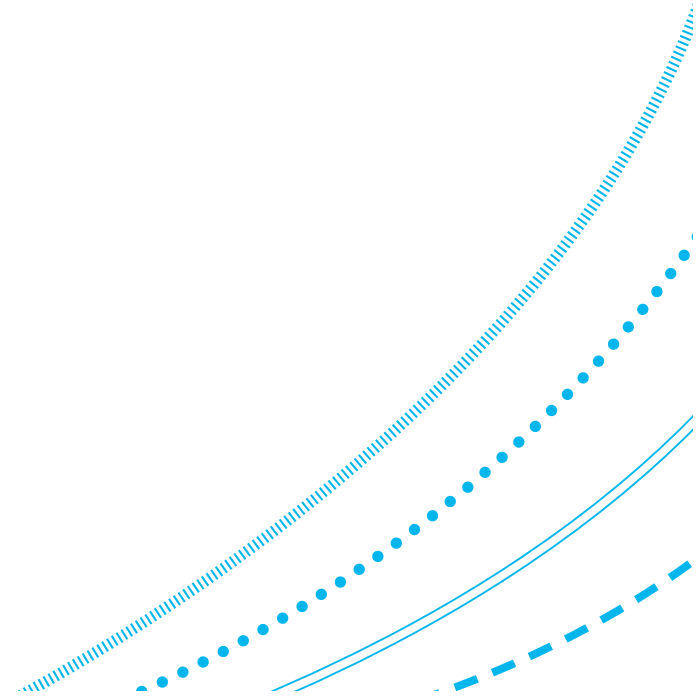
However, note that for a separate model where we compute the size from the price, the price (our model's input) would be the IV, and size (our model's output) would be the DV.



Prediction Vs Interpretation

Within supervised learning, once we have trained our model, there are two common goals that we might want to use our model to achieve:

Prediction and **Interpretation**.

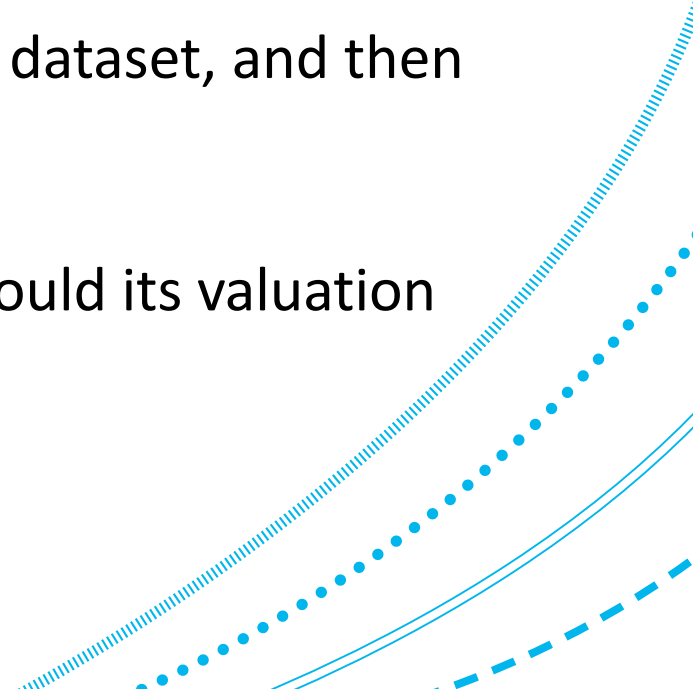


Prediction

In a 'typical' ML context, we might want to determine how well our model can be used to **predict** new (unseen) data.

We might first *train* a model, *test* it with a withheld dataset, and then *deploy* it to predict new data.

> E.g. if we have a house with 1,500 sq. ft., what should its valuation be?

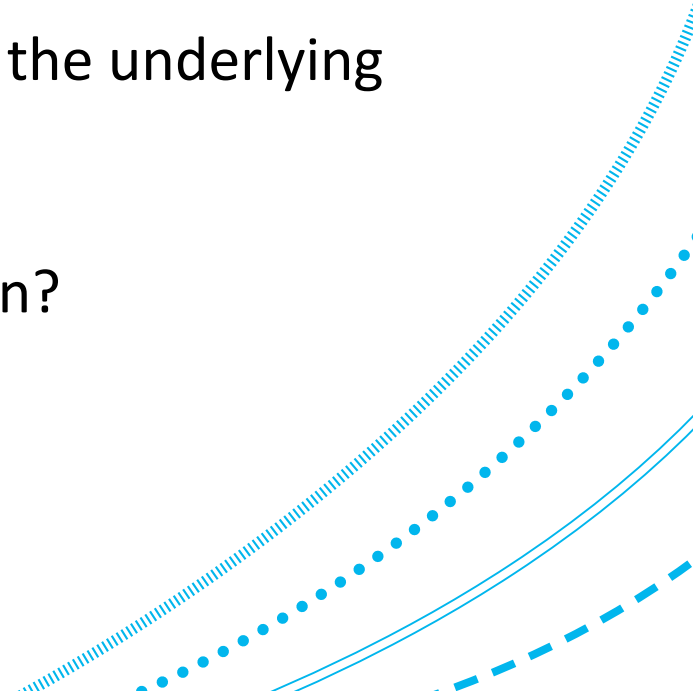


Interpretation

However, when performing data analysis, we might be more interested in what we can **infer** from the model.

In other words, we might want to understand what the underlying relationships within the data can teach us.

> E.g. How does a house's size influence its valuation?

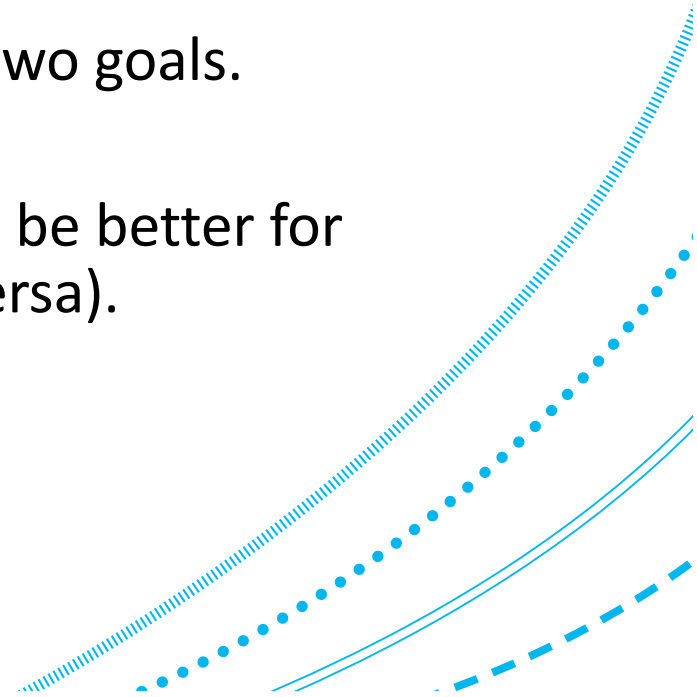


Prediction Vs Interpretation

Note that in practice, we might want **both** prediction and interpretation. Both are valid end-goals to have.

However, there is often a trade-off between these two goals.

Certain supervised learning algorithms will typically be better for prediction and worse for interpretation (and vice versa).

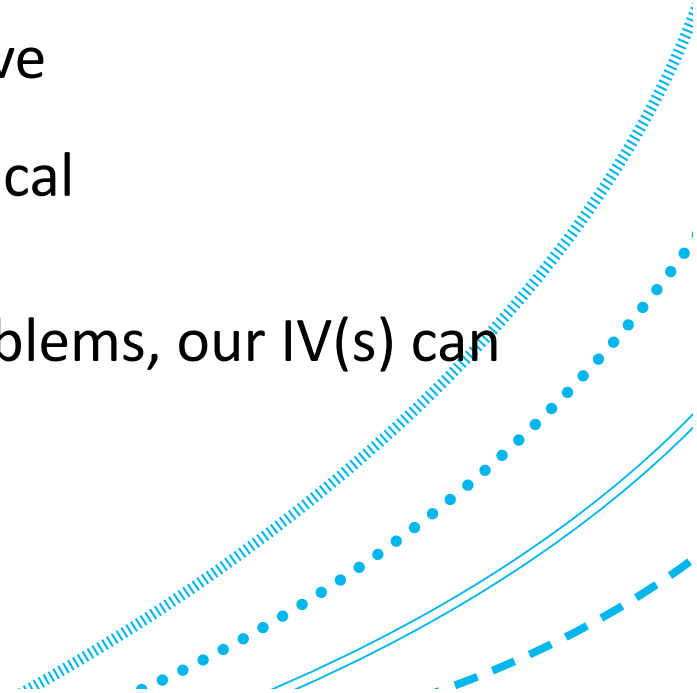


Regression Vs Classification Problems

When we discuss supervised learning, there are two broad types of problems that tend to come up:

- **Regression problems**, where our DV is quantitative
- **Classification problems**, where our DV is categorical

Note that for both regression and classification problems, our IV(s) can be quantitative and/or categorical.



Regression Analysis

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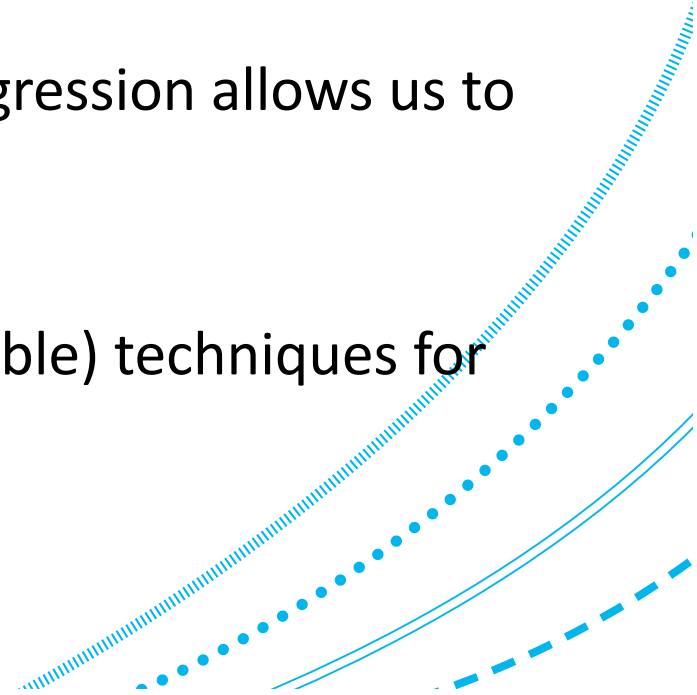


Regression Analysis

Regression analysis refers to a **set** of statistical methods for quantifying relationships between variables.

In this way, it is similar to correlation – however, regression allows us to do much more, above and beyond correlation!

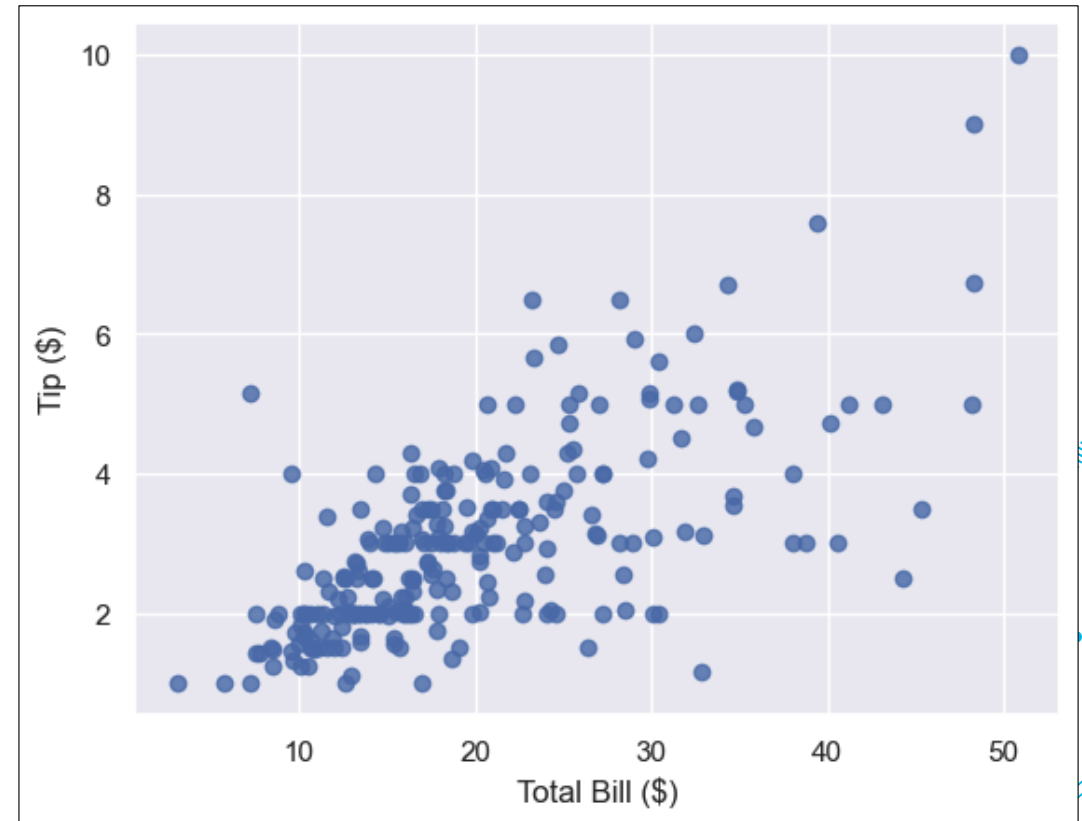
They are some of the simplest (and most interpretable) techniques for supervised learning.



Regression Analysis

Let's go back to some of our correlation examples.

Example: Do larger bills result in larger tips?

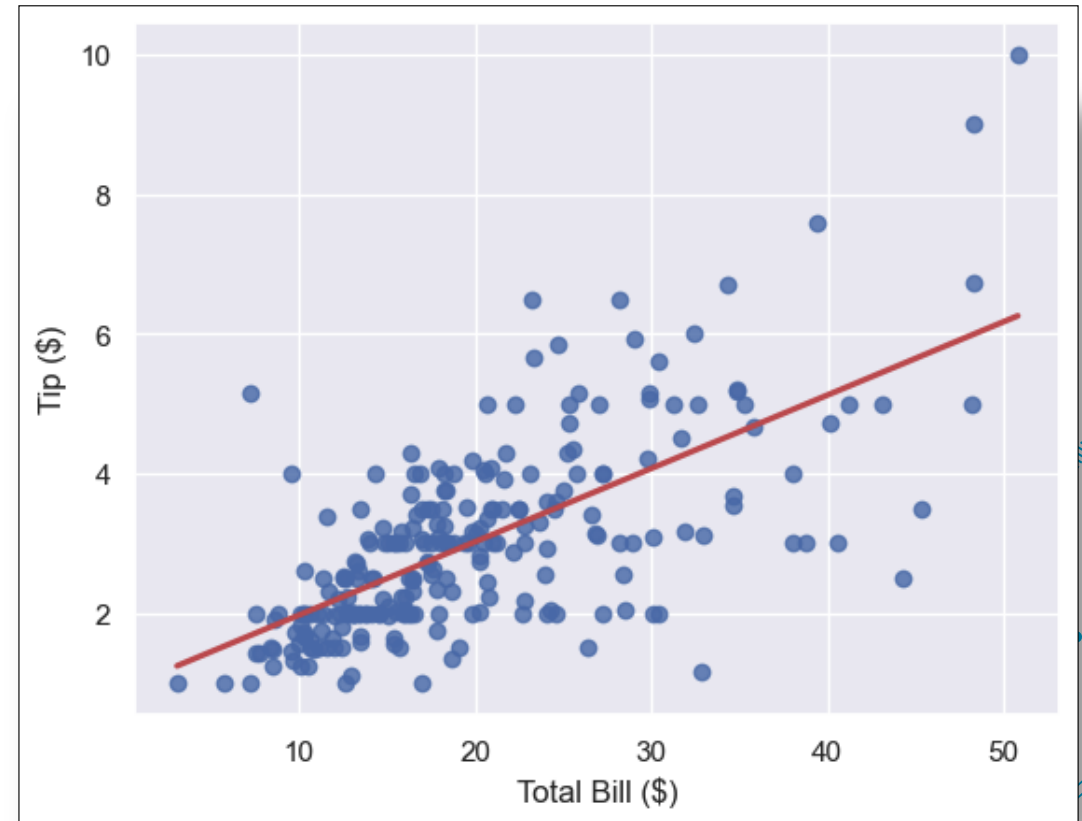


Regression Analysis

In previous examples, we used the 'line of best fit' to visualise our correlations.

This line is a regression model. It estimates the function that most closely fits the data.

Here, we have one predictor (IV).

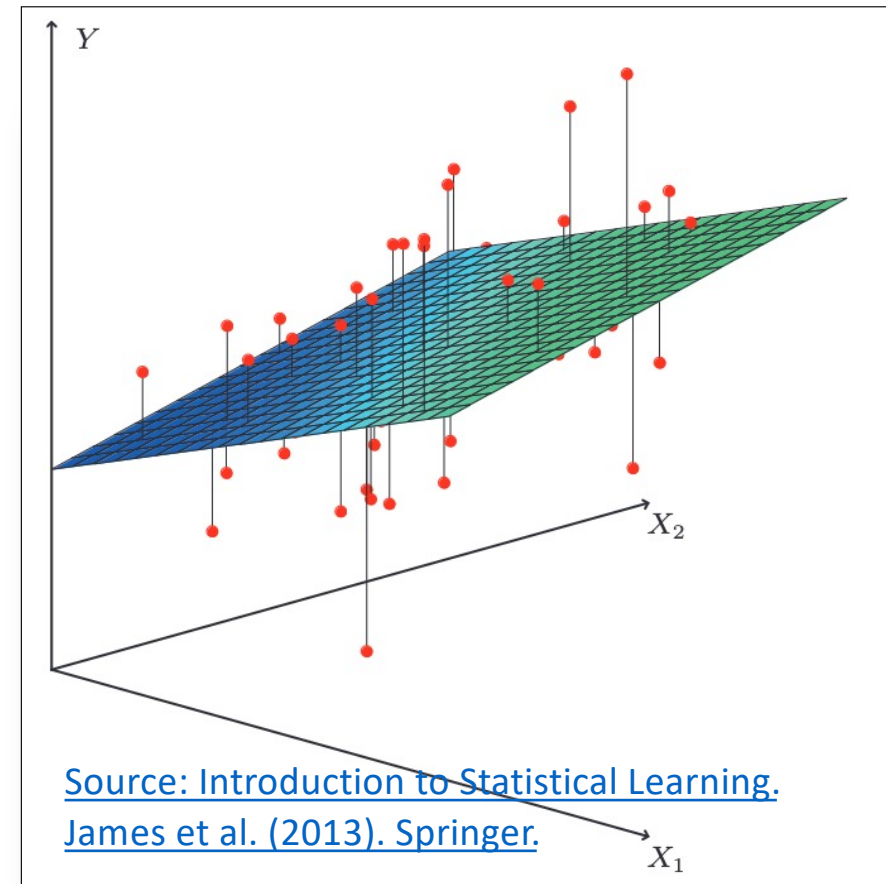


Regression Analysis

But regression allows us to have as many as we want.

Here, we now have 2 IVs (X_1 and X_2), and instead of a line, we are now using a plane.

But we could have hundreds of IVs if we wanted to.



Regression Models: Under the Hood

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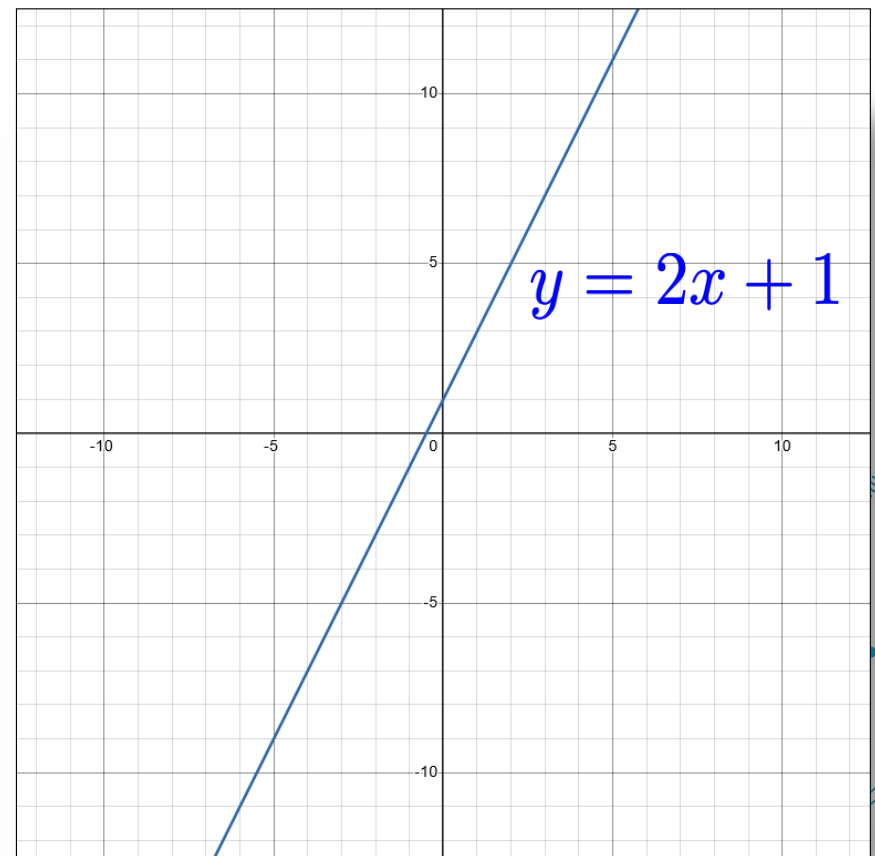
Regression Model

To understand what these regression models are, recall that the equation of a line:

$$y = mx + c$$

With one IV, the formula for regression is basically the same:

$$y \approx \beta_0 + \beta_1 x_1$$

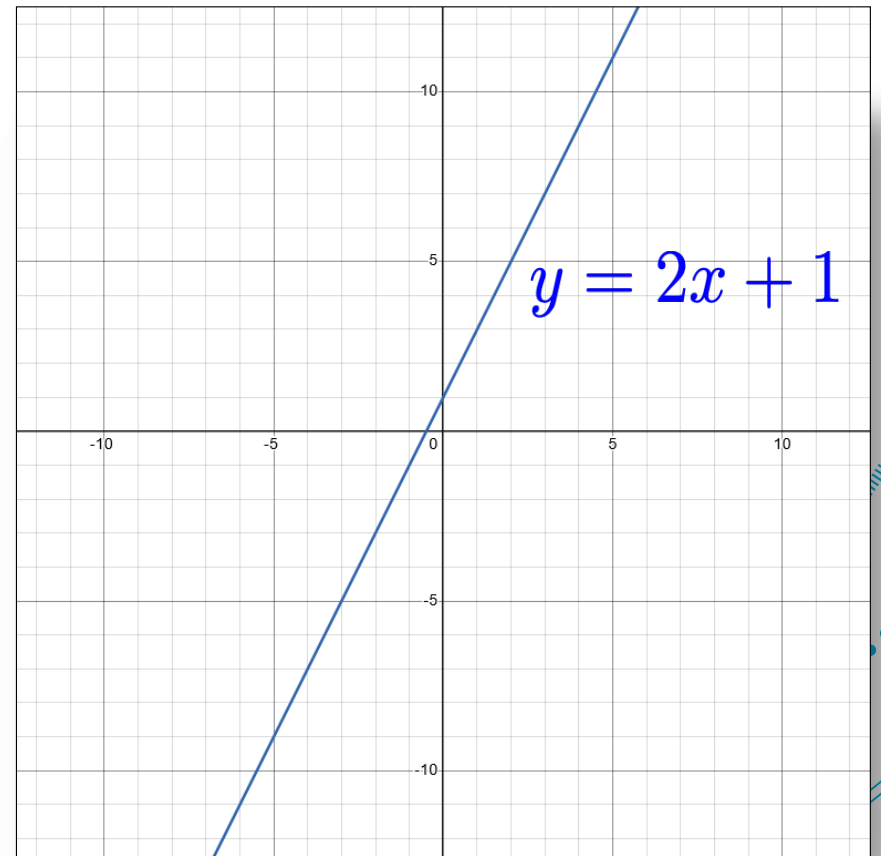


Regression Model

$$y \approx \beta_0 + \beta_1 x_1$$

Here:

- y is our DV
- x_1 is our IV
- β_0 is our y intercept
- β_1 is our gradient for x_1



Regression Model

Intercept: \$0.92

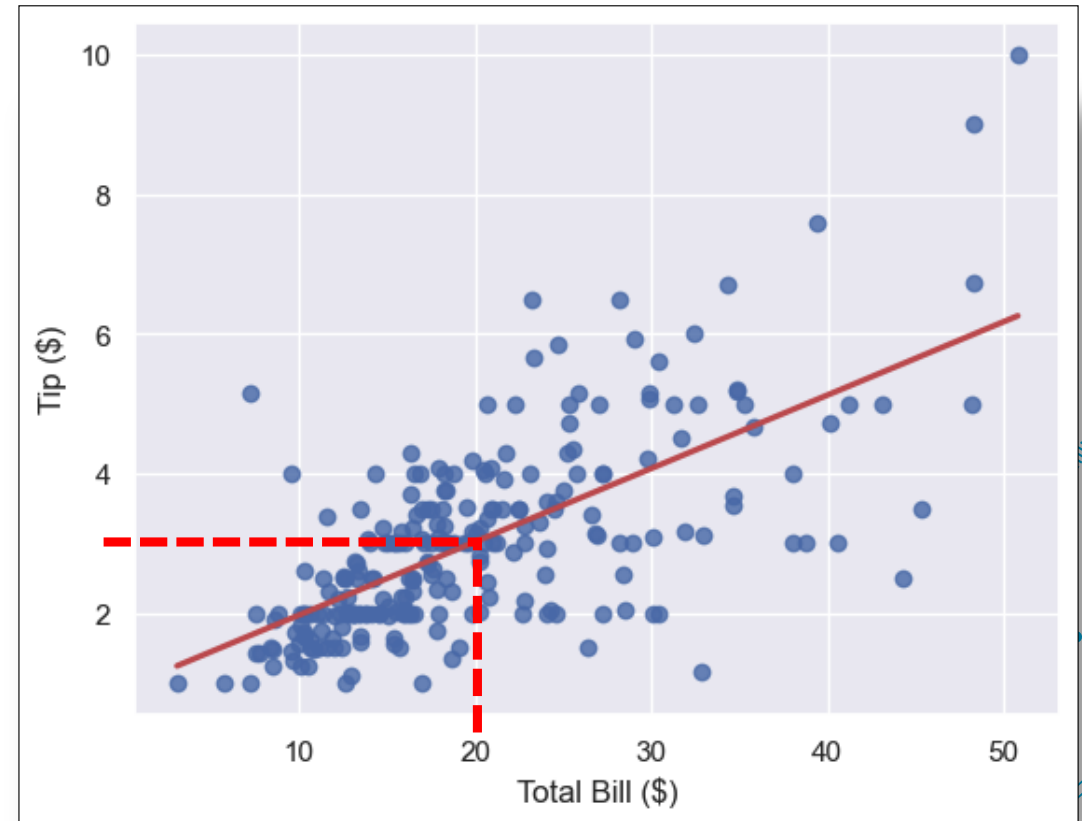
Gradient: 0.105

If we have a total bill of \$20, our tip will estimate to:

$$> y \approx \beta_0 + \beta_1 x_1$$

$$> y \approx 0.92 + (0.105 * 20)$$

$$> y \approx \$3.02$$



Evaluating a Regression Model

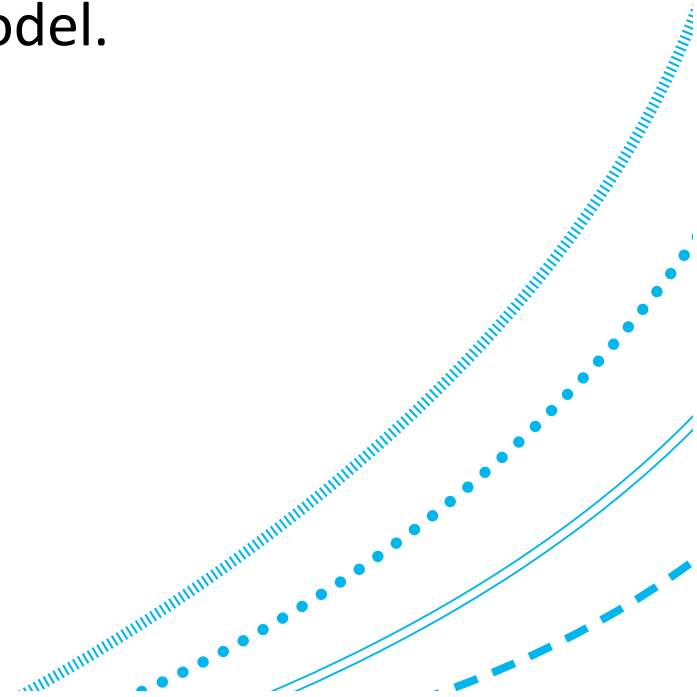
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Evaluating a Regression Model

We will often want to know how well our model 'fits' the data.

There are lots of ways of evaluating a regression model.

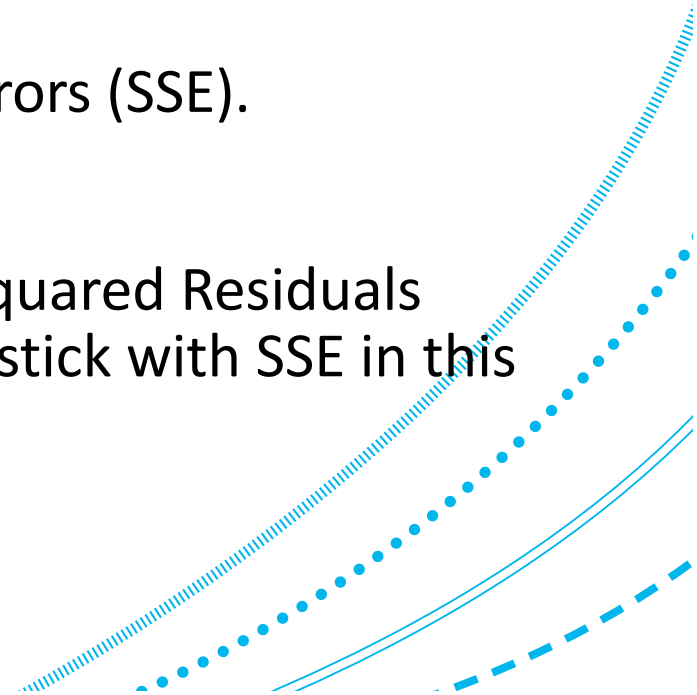


Sum of Squared Errors (SSE)

The coefficients in a linear regression models will often be estimated using the 'Ordinary Least Squares' (OLS) method.

Here, the aim is to minimise the Sum of Squared Errors (SSE).

Note that SSE is sometimes referred to as Sum of Squared Residuals (SSR), and Residual Sum of Squares (RSS), but we'll stick with SSE in this course.

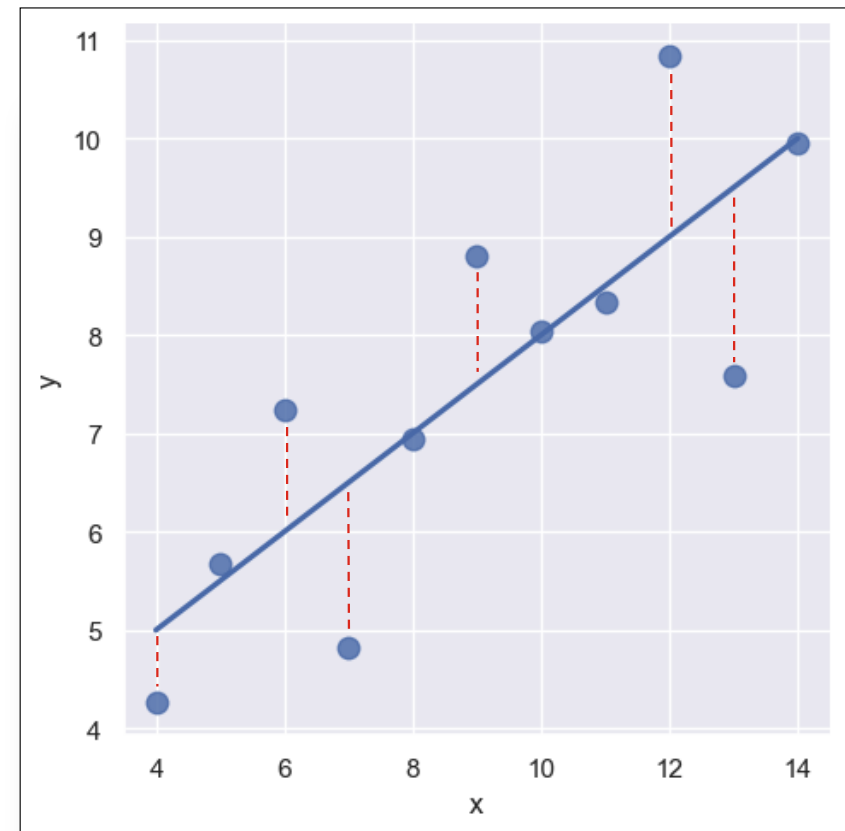


Sum of Squared Errors (SSE)

For each data point, we square the difference between the actual & estimated values.

Then, we add these up:

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



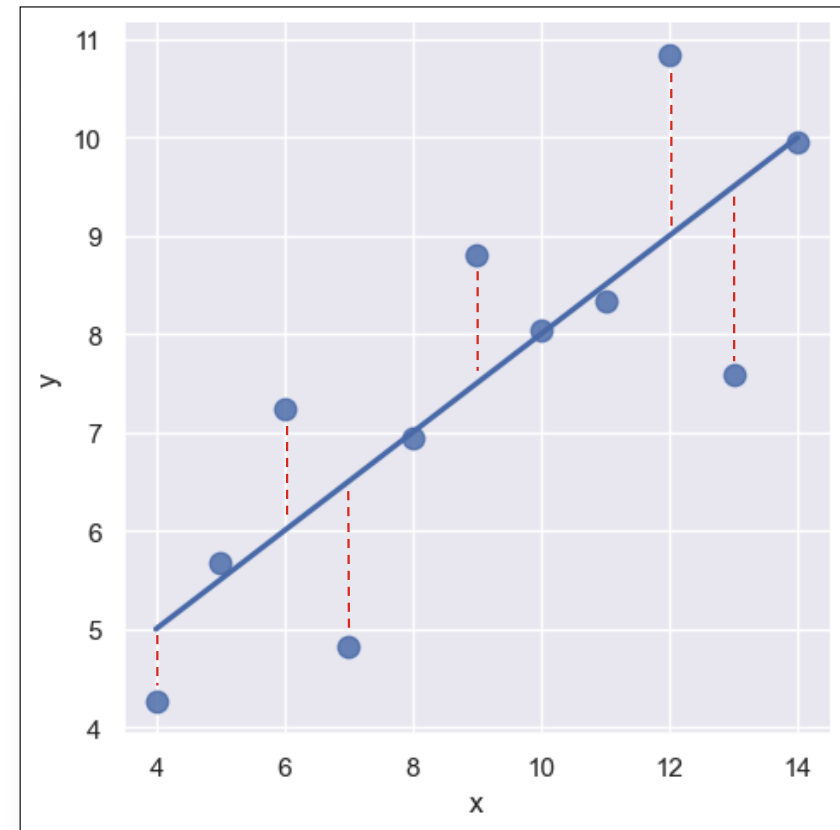
Sum of Squared Errors (SSE)

Sum of Squared Errors

↓ ↘ ↙

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

A small SSE represents a close fit.

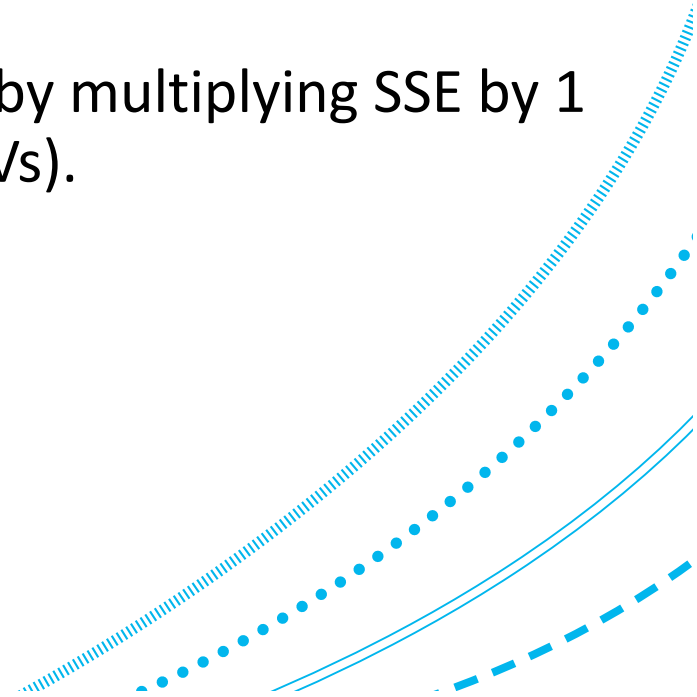


Mean Squared Error (MSE)

SSE is good for comparing different regression models against a dataset, but the SSE value will increase as n increases.

So instead, we can use Mean Squared Error (MSE), by multiplying SSE by $1 / (n - p - 1)$, where p is the number of predictors (IVs).

$$MSE = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

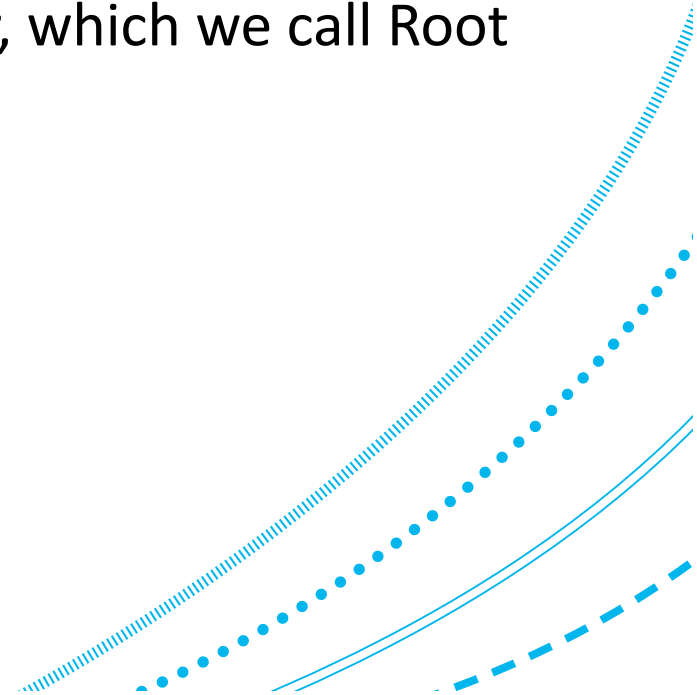


Root Mean Squared Error (RMSE)

So, MSE represents the average **squared** error.

If we find the root of this, we find the average error, which we call Root Mean Squared Error (RMSE), or Standard Error (SE)

$$RMSE = \sqrt{\frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



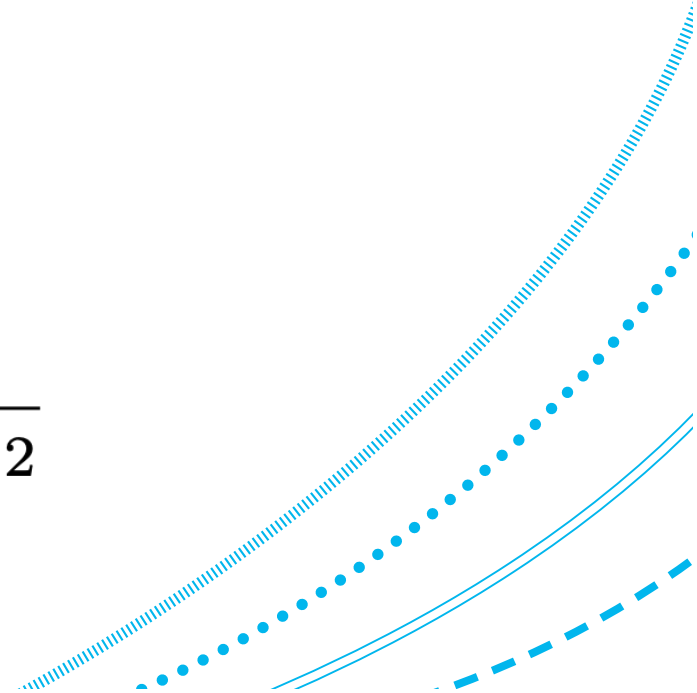
Evaluating a Regression Model

In other words, these three metrics are iterations of one another.

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



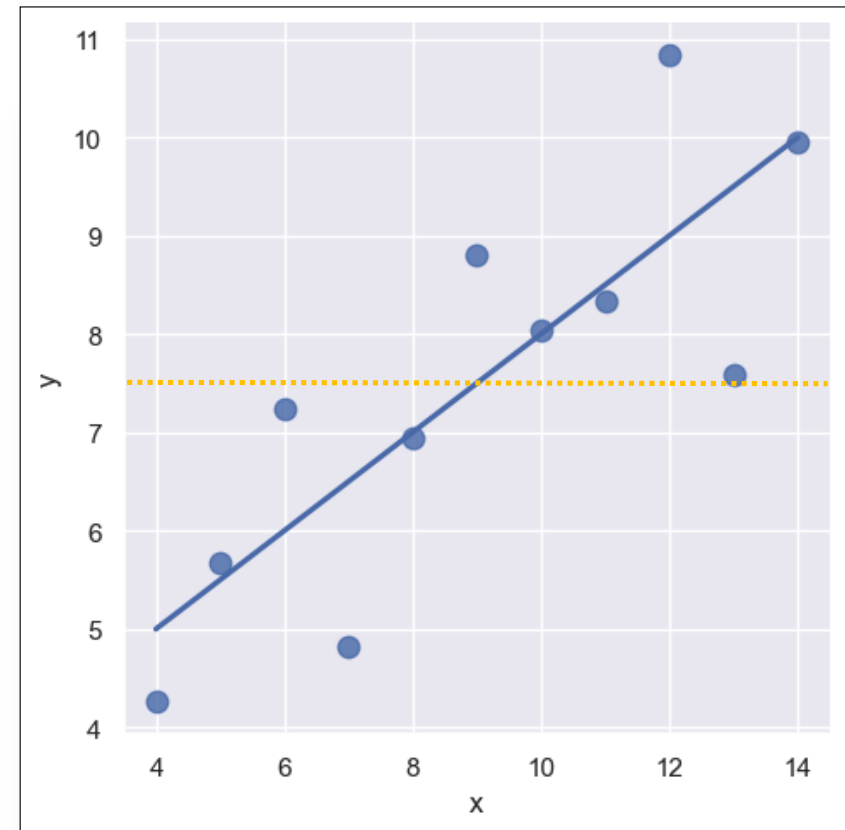
R² Value

Another way to evaluate our model is by comparing it to just using the mean.

Here:

- Blue line: $y \approx \beta_0 + \beta_1 x_1$

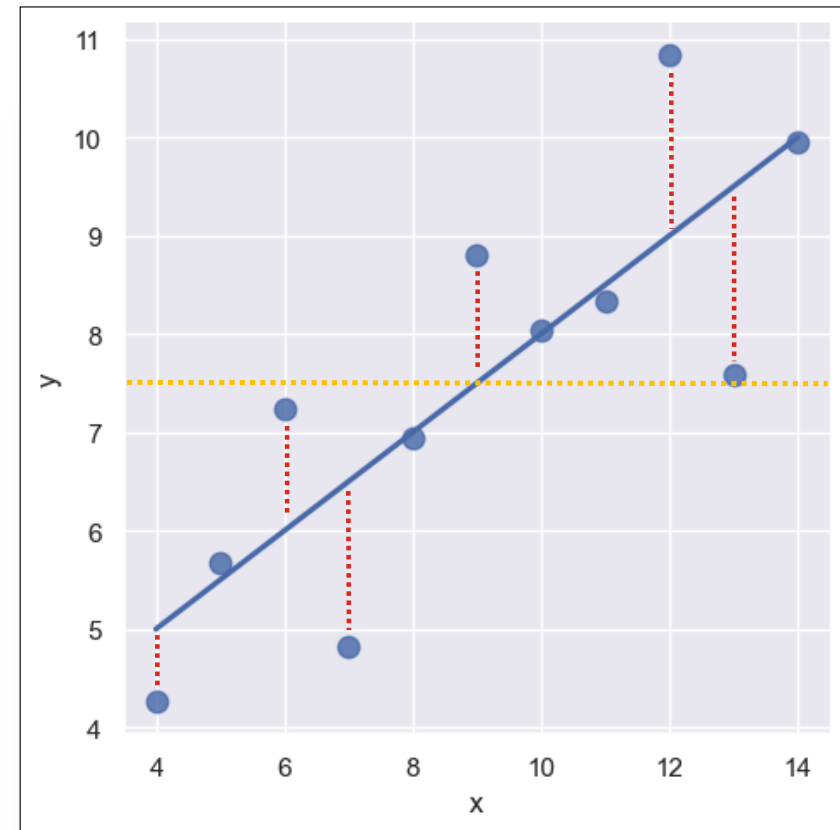
- Yellow line: $y \approx \frac{1}{n} \sum_{i=1}^n x_i$



R² Value

Recall the SSE calculation:

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



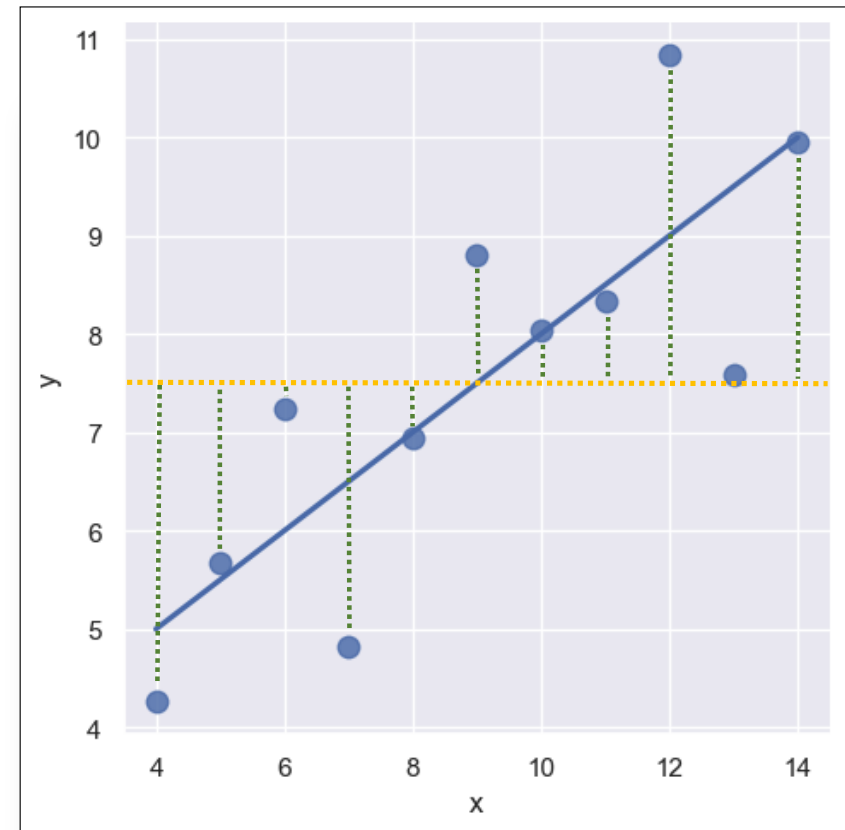
R² Value

Recall the SSE calculation:

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Let's say we calculate the Sum of Squares Total (SST), using the mean instead of $\hat{y}$

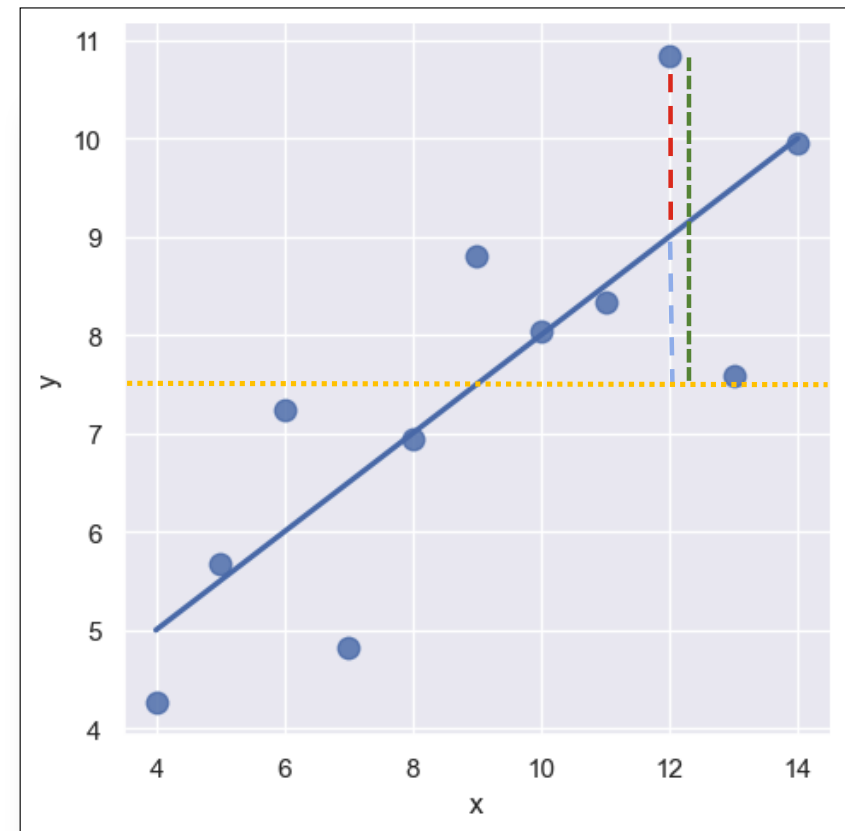
$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$



R² Value

One way to think of this is that we're differentiating between:

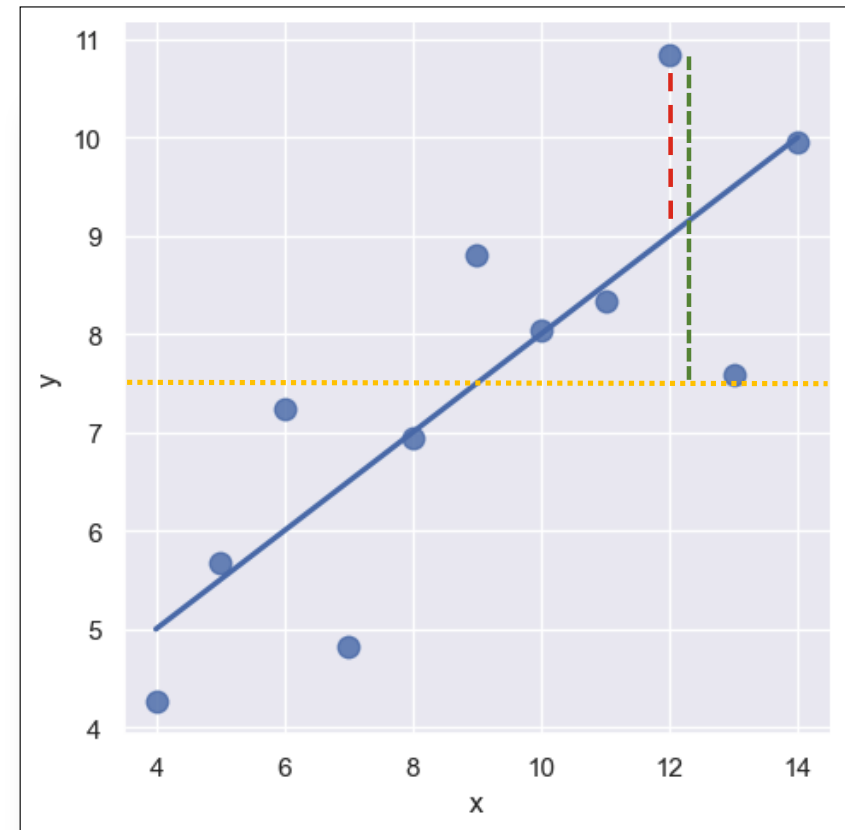
- variance that is **explained** by our model (.....),
- variance that is **unexplained** by our model (.....),
- and the total variance (.....)



R² Value

R² tells us what proportion of the total variance is explained by our regression model

$$R^2 = 1 - \frac{SSE}{SST}$$
$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

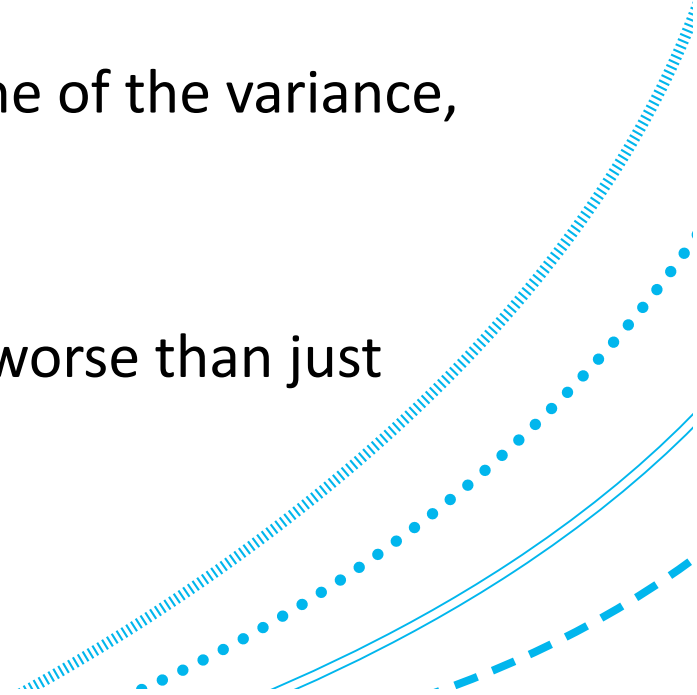


R² Value

An R² value of 1 means that our model perfectly explains all of the variance within the data.

An R² value of 0 means that our model explains none of the variance, and is no better than just using the mean.

An R² value of < 0 means that our model performs worse than just using the mean.

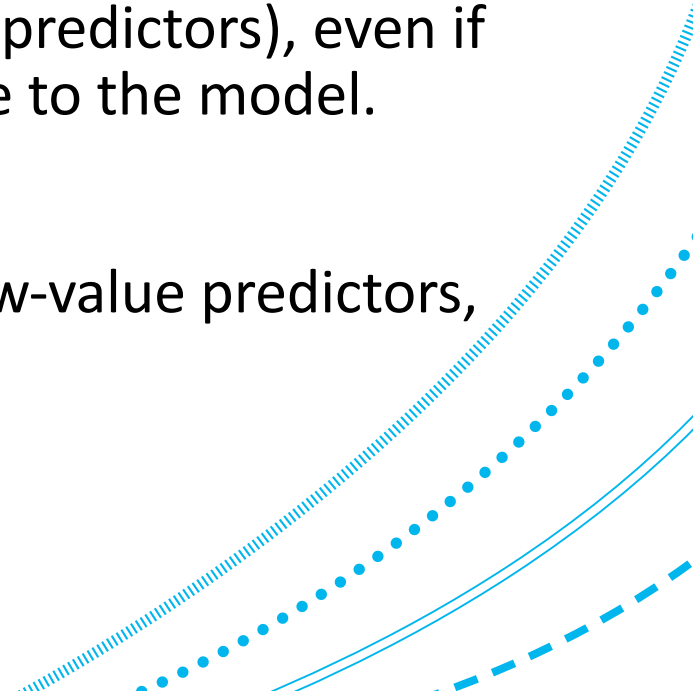


Limitations of R^2

However, note that the R^2 value has a significant weakness;

The R^2 value **never decreases** as you add more IVs (predictors), even if these predictors don't add any new predictive value to the model.

As a result, as we keep increasing the number of low-value predictors, our R^2 value will become inflated by collinearity.

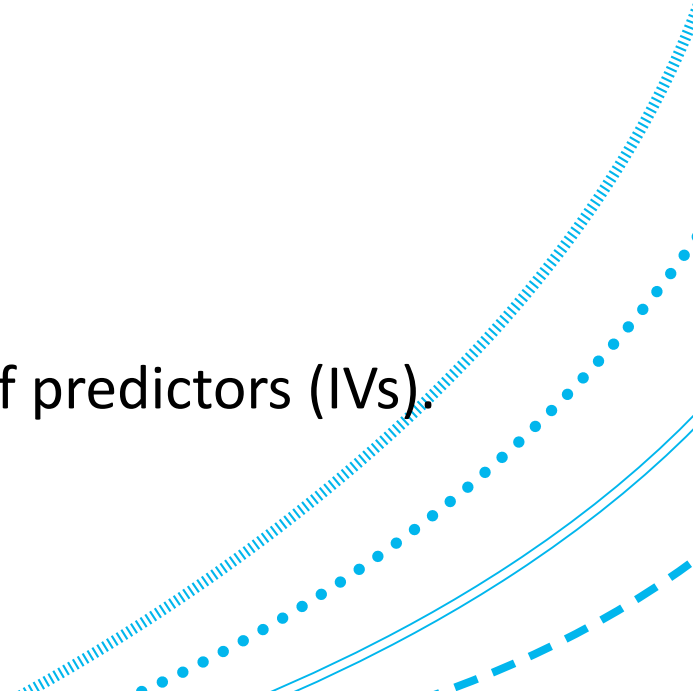


Adjusted R^2

To address this issue, and to penalise the addition of low-value predictors, we can use a slightly adjusted metric known as Adjusted R^2 :

$$R^2_{\text{adj}} = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where n = number of data points and p = number of predictors (IVs).

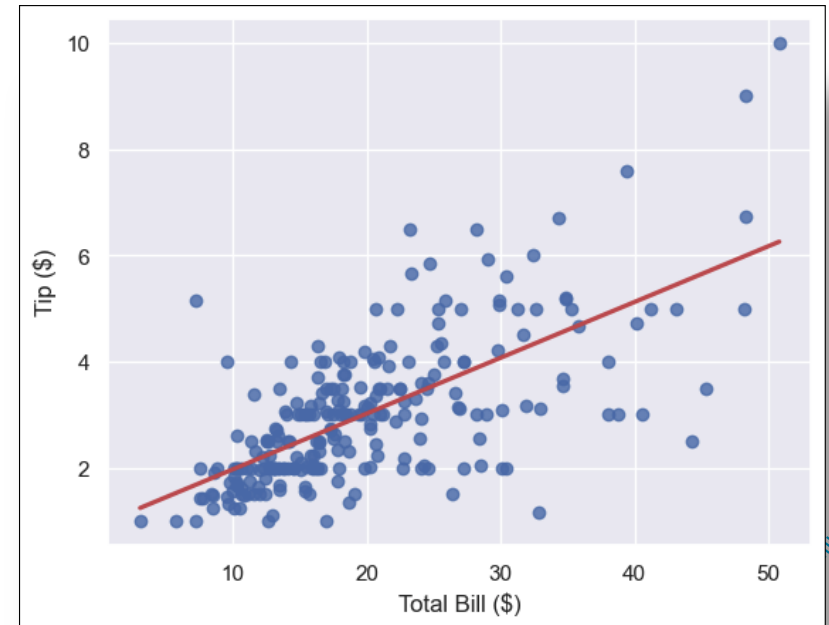


R^2 / Adjusted R^2 Vs RMSE

So, to clarify:

R^2 / Adjusted R^2 tell us **what proportion of the total variance is explained by our regression model.**

RMSE tells us **the average error between predicted and actual values** (in units of y).



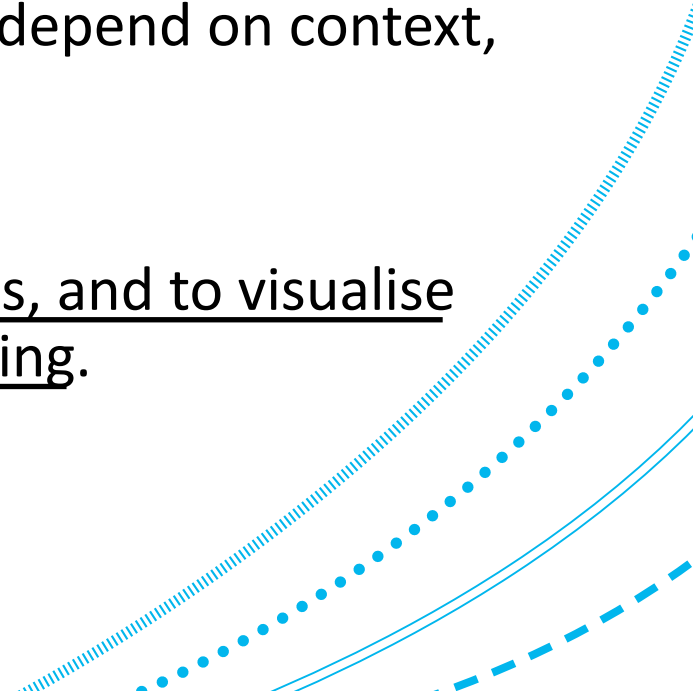
Intercept: \$0.92
Gradient: 0.105
RMSE: \$0.007
 R^2 : 0.457

R^2 / Adjusted R^2 Vs RMSE

Both metrics tell us something useful about the model.

But keep in mind that evaluation metrics will often depend on context, and any one metric may not tell the whole story.

Therefore, it is important to look at different metrics, and to visualise data (where possible) to gain a broader understanding.



What Can We Do With Regression?

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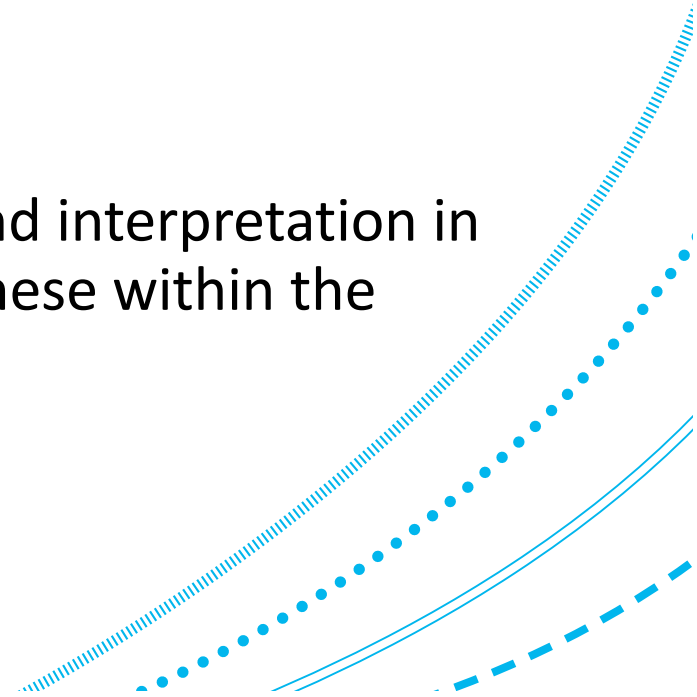


What Can We Do With Regression

So we've trained a regression model using a dataset – great!

But what can we do with this regression model?

We've already differentiated between prediction and interpretation in supervised learning, so let's demonstrate each of these within the context of regression.



Predicting With A Regression Model

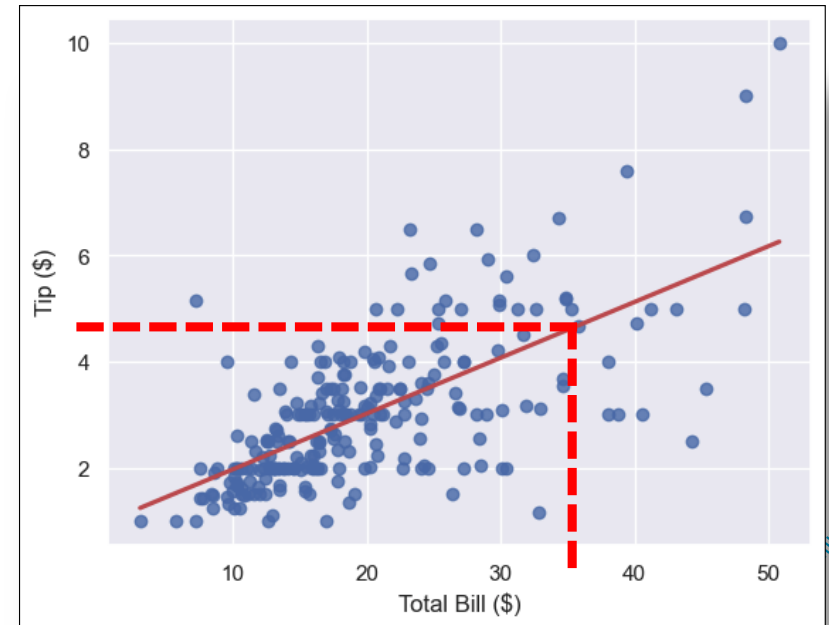
Prediction: We can use X values to predict new Y values.

If we have a total bill of \$35, our tip will estimate to:

$$> y \approx \beta_0 + \beta_1 x_1$$

$$> y \approx 0.92 + (0.105 * 35)$$

$$> y \approx \$4.60$$



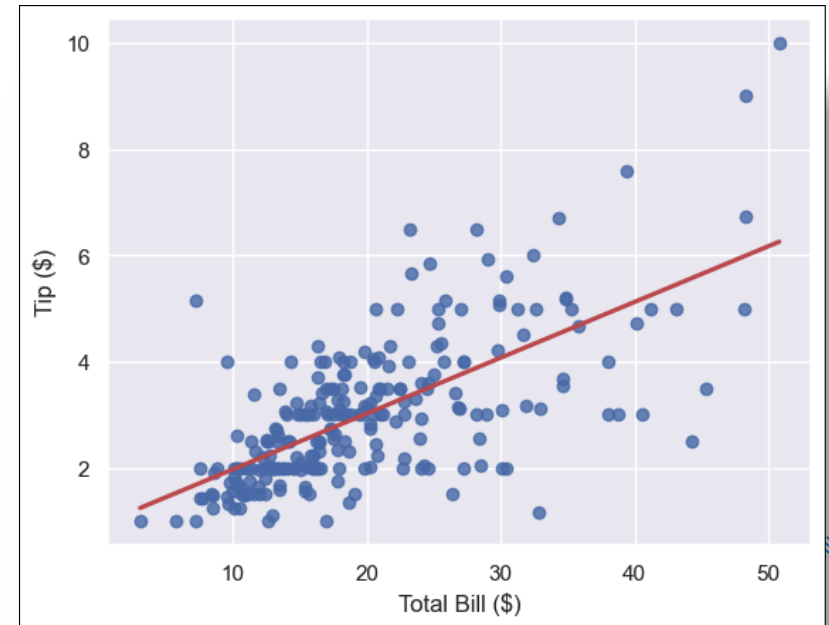
Intercept: \$0.92
Gradient: 0.105
RMSE: \$0.007
 R^2 : 0.457

Interpreting A Regression Model

Interpretation: We can use X values to understand how they influence the Y value.

For each \$ the total bill increases, the tip tends to increase around \$0.105. This is in addition to a constant value of \$0.92 (our intercept).

Basically, the tip is approximately 10.5% of the total bill plus an additional \$0.92.



Intercept:	\$0.92
Gradient:	0.105
RMSE:	\$0.007
R ² :	0.457

Interpreting A Regression Model

Note that in situations where we have several variables, interpretation becomes **very** useful.

We can use the regression output to understand how each of our independent variables interact with our dependent variable.

OLS Regression Results

Dep. Variable:	sales	R-squared:	0.278
Model:	OLS	Adj. R-squared:	0.275
Method:	Least Squares	F-statistic:	127.0
Date:	Sun, 11 Nov 2018	Prob (F-statistic):	0.00
Time:	14:48:08	Log-Likelihood:	-27899.
No. Observations:	5967	AIC:	5.584e+04
Df Residuals:	5948	BIC:	5.596e+04
Df Model:	18		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	7.4966	6.575	1.140	0.254	-5.394	20.387
weekend[T.Y]	12.4705	0.757	16.472	0.000	10.986	13.955
holiday[T.Y]	12.5009	1.911	6.542	0.000	8.755	16.247
prod_function[T.EyeCare]	-17.0896	1.738	-9.835	0.000	-20.496	-13.683
prod_function[T.Moisturisers]	-4.2329	1.081	-3.914	0.000	-6.353	-2.113
prod_function[T.Other]	-1.4569	1.600	-0.911	0.363	-4.594	1.680
prod_function[T.Toner]	-8.2722	1.626	-5.089	0.000	-11.459	-5.085
prod_function[T.Treatments]	-9.7475	1.246	-7.820	0.000	-12.191	-7.304
prod_regime[T.HERB]	14.5523	1.942	7.495	0.000	10.746	18.359
prod_regime[T.OCEAN]	1.0748	1.343	0.800	0.424	-1.558	3.708
prod_regime[T.POME]	1.7154	1.896	0.905	0.366	-2.002	5.433
prod_regime[T.TEA]	16.3640	1.141	14.337	0.000	14.127	18.601
prod_regime[T.VITA]	6.6399	1.106	6.003	0.000	4.472	8.808
prod_regime[T.WHITE]	23.1618	2.031	11.405	0.000	19.181	27.143
prod_regime[T.YOUTH]	47.4428	2.708	17.518	0.000	42.134	52.752
month	2.5016	0.683	3.660	0.000	1.162	3.841
prod_listprice	-7.49e-05	7.77e-06	-9.642	0.000	-9.01e-05	-5.97e-05
discount	0.4146	0.035	12.005	0.000	0.347	0.482
stockout	-32.6950	1.124	-29.086	0.000	-34.899	-30.491

Omnibus:	5081.500	Durbin-Watson:	1.821
Prob(Omnibus):	0.000	Jarque-Bera (JB):	201849.324
Skew:	3.904	Prob(JB):	0.00
Kurtosis:	30.403	Cond. No.	4.89e+06

Other Types of Regression

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Multiple Linear Regression

Again, we will often be interested in understanding the effects of more than one IV (predictor) on our DV.

This is often referred to as 'multiple linear regression', and mathematically, the idea is the same:

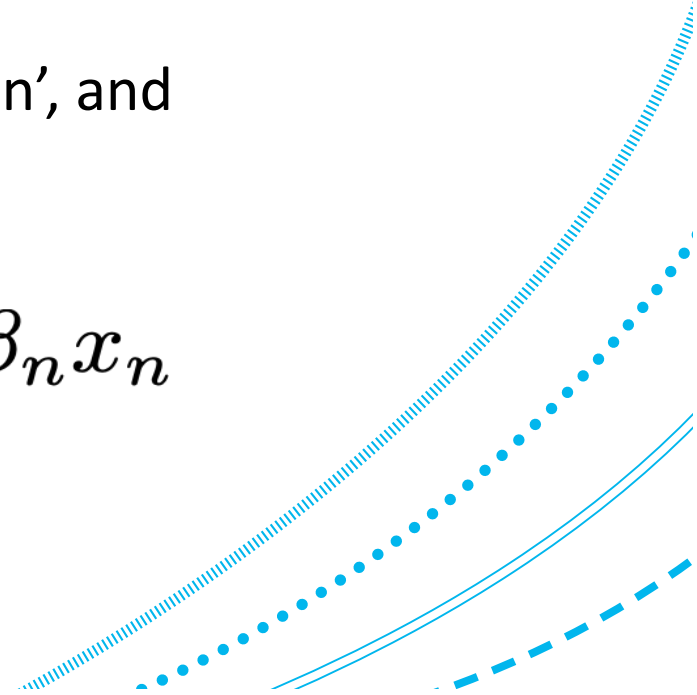
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Intercept

IV 1

IV 2

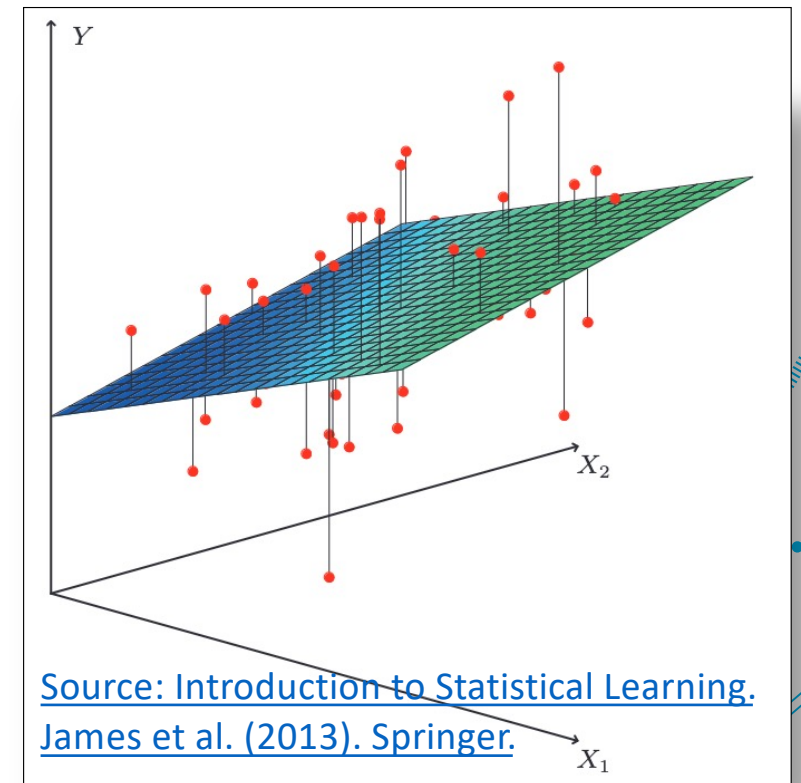
And so on...



Multiple Linear Regression

As the number of IVs increase, it becomes a little more complicated to conceptualise visually.

But for 2 IVs (X_1 & X_2), instead of fitting a line, we can just think of fitting a 'plane of best fit'.

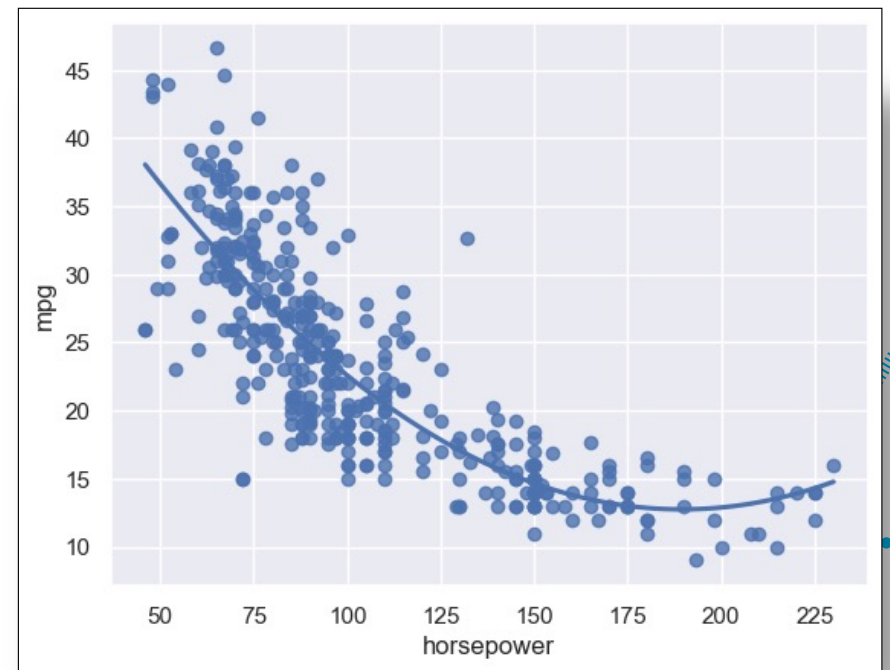


Non-Linear Regression

We can also build regression models with non-linear functions (e.g. polynomials).

This can result in a better 'fit' when relationships are non-linear.

However, interpretability becomes more of a challenge.

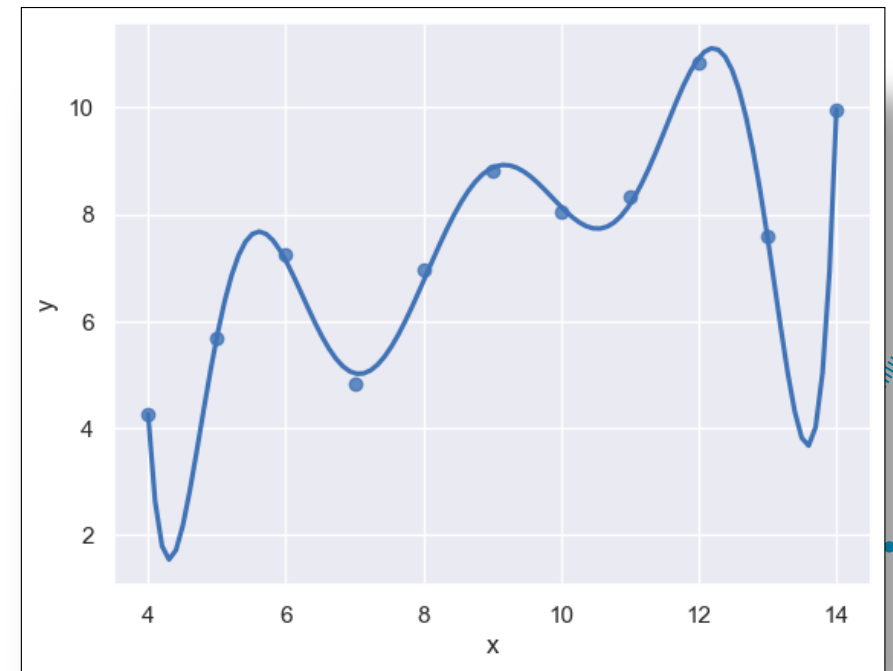


Non-Linear Regression

It also risks overfitting to the data.

This is especially the case when the model is made too complex.

Again, Python won't stop you from making the model overfit the data!



Limitations of 'Ordinary' Linear Regression

In short, regression is a very simple, interpretable, and useful method for analysing quantitative DVs.

But what if we want to predict a *categorical* variable?

Next up, we'll introduce *Logistic Regression*, which will allow us to do this.

